Explaining Citizen Satisfaction: Searching for Alternatives to the Expectation-Disconfirmation Theory

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I. Introduction

Citizen satisfaction is used more and more by public managers (Bouckaert *et al.* 2005; Miller *et al.* 2009; Poister and Thomas 2011) as well as in research. In organizations where outcomes are difficult to measure, user satisfaction is used instead of more objective measures of performance. Others think of citizen satisfaction as an objective in its own right. This widespread use of satisfaction surveys as a measure of service performance has generated an obvious interest in what drives satisfaction (Charbonneau and Van Ryzin 2012; James 2009, 2011; Morgeson 2012; Van Ryzin 2004; 2006) The predominant theory of citizen satisfaction is the Expectation-Disconfirmation Theory suggesting that satisfaction is a reflection of the difference between the expected service level and the perceived service level or performance. This is basically a theory of a computational kind of rationality that suggests that agents evaluate a situation by comparing their preferences (expectations) to information about the state of the world (perceived performance). Based on this information, they make a judgment (satisfaction).

In this study we find that citizens respond in much less rational ways to satisfaction surveys—in ways that are better explained by a more associative than rational mode of thinking (Kahneman 2011). First, we find that citizens' perceived performance of something situated in the past is amendable to political statements about current and future budget situations. Prior research has also found that perceived levels of governmental performance are influenced by information about the state of the world (performance information or pictures of street-cleanliness). This has not been interpreted as evidence against the Expectation-Disconfirmation Theory, though, because it has been tested on either hypothetical examples (Van Ryzin 2013) or on service at a very general (abstract) level on which citizens are not expected to have any firm experiences themselves (James 2011). We use a randomized field experiment (study 1) to test how real messages from local governments about current and future budgets affect how citizens (parents) perceive their experience with services with which they do have personal experience (the quality of their childrens' school). To bolster our findings we replicate them on both more

abstract and more concrete service areas, and on real political messages that both suggest budget cutbacks as well as status quo.

This result could be explained by an associative mode of thinking in which political messages about budget constraints primes for associations to low performance and bad quality (Kahneman 2011). To further test the relevance of this theory, we secondly move on to test how satisfaction is related to real experiences. If satisfaction is to be an appropriate measure of performance (on some dimensions), it should be a reflection of citizens' experiences with the public services (given some level of expectations). This would be the claim by the Expectation-Disconfirmation Theory. However, in a second experiment (Study 2) we show, as expected by dual process-theories of reflective and associative or intuitive thinking (Kahneman and Frederick 2002), that satisfaction is much stronger correlated to experience if this experience is invoked in the mind just before answering the survey question about satisfaction, than if they are invoked immediately after answering the question. Apparently, politicians and public managers can expect quite different performance evaluations from citizens if they are reminded of their experiences just before evaluating, than if they are not. Experiences that according to our first experiment also are amendable to irrelevant information.

In the next section we discuss the existing evidence of the Expectation-Disconfirmation Theory and contrast it to general research on reflective and associative modes of thinking. Then we present the research design that uses experiments embedded in satisfaction surveys among parents of school children in the City of Aarhus in Denmark to test basic assumptions of the Expectation-Disconfirmation Theory. After having presented the results, we conclude by discussing how this study brings our understanding some way beyond the hitherto dominating theory of a rational relationship between expectations, experiences and satisfaction. We do not claim that satisfaction is unrelated to personal experiences with performance. But we argue that it is not a relationship as stringent as suggested by the theory, and that the associative mode of thinking is important to understand for both practitioners and researchers that use satisfaction surveys to measure organizational performance.

II. Theory

The Expectations of the Expectation-Disconfirmation Theory

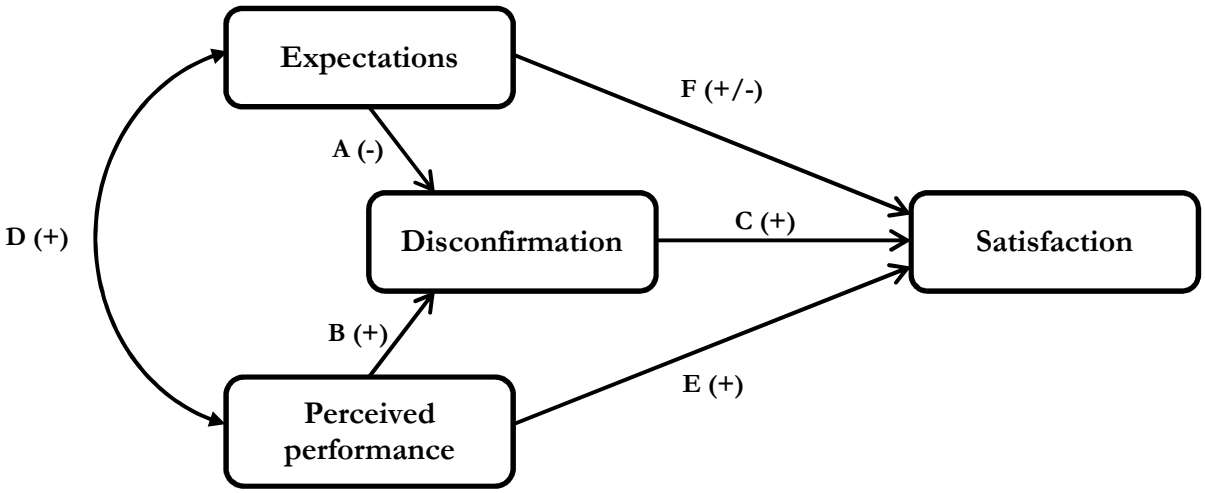
Developed and tested in the private sector consumer satisfaction research, the Expectation-Disconfirmation Theory (EDT) has a long history (Cardozo 1965; Oliver 1980, 1997; Tse and Wilton 1988; Yi 1990). Citizen satisfaction with services delivered by the public sector surfaced during the last decade, and this literature has been quick to adopt the EDT as well (Van Ryzin 2004, 2006; Roch and Poister 2006; James 2009, 2011a, 2011b; Poister and Thomas 2011; Charbonneau and Van Ryzin 2012; Morgeson 2012).

The EDT states that if the consumer's or citizen's prior expectations are disconfirmed by the perceived performance of the product or service, they will be dissatisfied. Conversely, if the expectations are confirmed by the perceived performance, or even exceeded, they will be satisfied (Cardozo 1965; Oliver 1980, 1997; Tse and Wilton 1988; Zeithaml *et al.* 1993).

Expectations are here seen as an anticipation of the performance of the service (or product) created from prior experiences with the service, word-of-mouth, communication in the media and so on. They are "*a frame of reference about which one makes a comparative judgement*" (Oliver 1980, p.460). This essentially builds on Helson's adaptation level theory in which any evaluation of a stimulus is made in comparison with an adapted standard. The adapted standard is created before the current evaluation and is hard to change (Helson 1948). The perceived performance of the service at any given time point is then contrasted with this adapted standard, resulting in the disconfirmation construct. Since the expectations are assumed created prior to the other constructs in the theoretical model and the perceived performance is assumed based on a recent experience with the service these constructs are considered exogenous in the literature (Van Ryzin 2004, p.436). Lastly, the satisfaction evaluation is made heavily influenced by the disconfirmation (Oliver 1980).

These theoretical relationships are illustrated in Figure 1.

Figure 1: The Expectation-Disconfirmation Theory



Note: Adopted from Van Ryzin (2013)

In Figure 1, expectations and perceived performance directly affect the disconfirmation construct (A and B). All else equal, higher perceived performance leads to a more positive disconfirmation, and higher expectations leads to less positive disconfirmation. Usually, disconfirmation is operationalized as either the difference between the expectations and the perceived performance items in a survey (subtractive disconfirmation) or directly as a single item (perceived disconfirmation) (Van Ryzin 2006). Disconfirmation is hypothesized to be positively related to satisfaction (C). These links are the basics of the EDT. The bidirected arrow between expectations and perceived performance (D) is usually found to be positive, but there seems to be no explicit theory behind this link and no agreement if the link is a basic part of the EDT (Van Ryzin 2006, p.601, 2013, p.4).

The direct link between perceived performance and satisfaction (E) is naturally positive association and usually a very strong one (Lyons *et al.* 1992). The last link (F) suggests either a positive or negative relationship between expectations and satisfaction. This is not an integrated part of EDT, although a negative effect of expectations on satisfaction would be in congruence with it (Poister and Thomas 2011). However, some applications have found a positive relationship, which is sometimes

referred to as expectation anchoring, where the initial expectations create an anchoring effect (Tversky and Kahneman 1974; Van Raaij 1989) on the satisfaction (Olshavsky and Miller 1972; Olson and Dover 1976; Oliver and DeSarbo 1988; Van Ryzin 2004).

The model is difficult to test empirically – especially on observational data. Since satisfaction is likely to affect subsequent expectations, an endogenous relationship between the two cannot be ruled out. Also the bidirected relationship between expectations and perceived performance (D) makes it difficult to isolate the effect of disconfirmation from separate effects of expectation or experience that work through one another. Only recently has experimental designs been used to test models of citizen satisfaction (James 2011a, 2011b; Van de Walle and Van Ryzin 2011; Van Ryzin 2013), but even in experimental designs it is difficult to identify causal mediation mechanisms between variables in the model (Imai *et al.* 2011). For instance, if an experimental treatment affects expectations, and the treatment also correlates with satisfaction, it is difficult to know if the effect of the treatment works through expectation or some other unobserved variable (or affects satisfaction directly).

However, a number of testable implications of the EDT can be derived. One implication concerns the expectations: An increase in expectations should – all else being equal (including no change in perceived performance) – result in lower satisfaction, because higher expectations without an increase in performance should generate disconfirmation and thereby dissatisfaction. This is not what Van Ryzin (2013) finds, though. He experimentally manipulates respondents' (an Internet panel) expectations by presenting them to two different hypothetical public statements by a city administrator. One informed citizens that budget cuts had to be made. The other that there would be no reduction in the quality of public services. The treatment had a significant effect on respondents' expectations, but there were no effect on satisfaction. This contradicts one of the basic implications of the EDT.[1]

Another implication of EDT concerns the perceived performance. If satisfaction surveys are related to the performance of public authorities in a systematic manner, the perceived performance should

[1] It could be that the treatment also affected the perceived performance. This test is not presented in the article.

be a reflection of real performance. This is the (computational) rational aspect of the theory: Citizens are expected to use their information about past performance and weigh it against their expectations to form a judgment about their satisfaction. In a study of the link between performance information and satisfaction, James (2011a) experimentally manipulates people's perceived performance by either giving them or not giving them information about the relatively high performance of their local council. James (2011a, p.402) notes that citizens may "not have much of an idea about the overall performance of a local public body only interacting with it on a case-by-case basis for a subset of services". If satisfaction should be a reflection of performance (holding expectations constant) it seems at odds with the model that performance can be effectively manipulated by the information provided. In this case the general performance of the public authority is a rather abstract concept, which may explain why the perception can be manipulated (Stipak 1979; Parks 1984), but the result raises a question about whether citizens' perceptions of performance can be manipulated at areas where they do have personal experience.

To test this question, we make two variations compared to James (2011a) in what we call Study 1. First we make the performance more concrete by asking about perceived performance on both a medium and low level of abstractness. The more concrete the evaluation the less the perceived performance should be malleable. Much of the satisfaction literature examine cases where respondents do not have experience with the service they are asked to evaluate (e.g. Kelly and Swindell 2003; Van Ryzin 2013). We test the question in a field experiment where citizens are asked about their satisfaction with services that they actually have experienced. That should also make satisfaction less malleable. Second, we do not provide information about past performance, as James (2011a) did, but about current and future budget constraints, as Van Ryzin (2013) did. Whereas information about past performance is relevant to perceived performance, information about current and future budget constraints should not affect experiences of past performance, according to EDT.

As the results from Study 1 will show, however, responses to the question about perceived performance are affected by these information treatments. That suggests that citizens do not respond to

satisfaction surveys in as (computationally) rational a way as suggested by the EDT. Rather it seems as if a more associative mode of thinking is taking place in the sense that the framing about current and future budget situations seem to prime for associations of budget restrictions and, relatedly, poor performance, which may explain why these framings affects the perception of experienced performance.

An Alternative Theory of Citizen Satisfaction

These two modes of thinking has been labeled by Stanovich and West (2000) as "System 1" (the associative or intuitive mode) and "System 2" (the rational or reflective mode) (for a thorough presentation and discussion, see Kahneman 2011). Research has consistently demonstrated how people use System 1 by using associations from both affective and cognitive feelings when making judgments about things such as life satisfaction. Affective feelings are valenced subjective experiences encompassing moods and emotions. Cognitive feelings include experiential states that accompany cognitive processes such as the ease with which information can be retrieved from memory (Schwarz and Clore 2007; for reviews of this research, see Greifeneder *et al.* 2011). Regarding the affective feelings, studies have found that people judge their life satisfaction as lower when they are in a bad mood – even when that mood is manipulated and therefore should not have an informational relevance for their judgment. This may be the explanation for the findings in study 1: The priming of citizens with information about budget constraints may set them in a bad mood that affects their perceived experiences with performance – even if the content value of the information is irrelevant for the judgment. We return to this when discussing the results.

The cognitive feelings relate, among other things, to the feeling of how easy it is to retrieve information from memory. It has been shown that by asking students about how many dates they have been on recently just before asking them about their life satisfaction, creates a much stronger association between dates and satisfaction than if the question about dates is asked after the question of life satisfaction (Strack *et al.* 1988). That suggests that when primed with the dates question, System 1

substitutes the broader question about life satisfaction in general with the narrower question of satisfaction with the romantic aspects of one's life (Kahneman 2011).

However, this relationship is possibly more complicated. A famous study shows that when people are asked to remember few versus many instances of previous assertive behaviours, they judge themselves as being *more* assertive when remembering few instances rather than many. This is explained by the fact that it is much easier to remember few than many instances. In other words, rather than relying on the content information – the number of instances remembered – people rely on the easiness of retrieval: If it is easy to recall some instances, I must be quite assertive (Schwarz *et al.* 1991).

Based on this strand of research, we suggest the relationship between citizens' perceived performance and their satisfaction with public services is not just moderated by their expectations, as suggested by EDT, but that citizens also rely on these kinds of affective and cognitive feelings when making judgments about their satisfaction.

III. Design and Data

In order to test the different assumptions underlying EDT we conducted two studies in the context of public schools in Denmark. Experiments have the advantage of being able to identify the causal effect of different treatments and rule out the possibilities of endogeneity (Mutz 2011). Since these problems potentially could plague the EDT, experimental tests of the theory seems the most obvious choice (Van Ryzin 2006, 2013; James 2009).

More specifically, we use data from two surveys sent out to parents of 2nd and 3rd grade level children in 29 schools in the City of Aarhus in the autumn of 2013 and in the spring 2014 respectively[2]. The schools self-selected into a larger research project and, as a part of this larger project, the two surveys were issued to the parents.

[2] At the time, there were in total 49 public schools in Aarhus

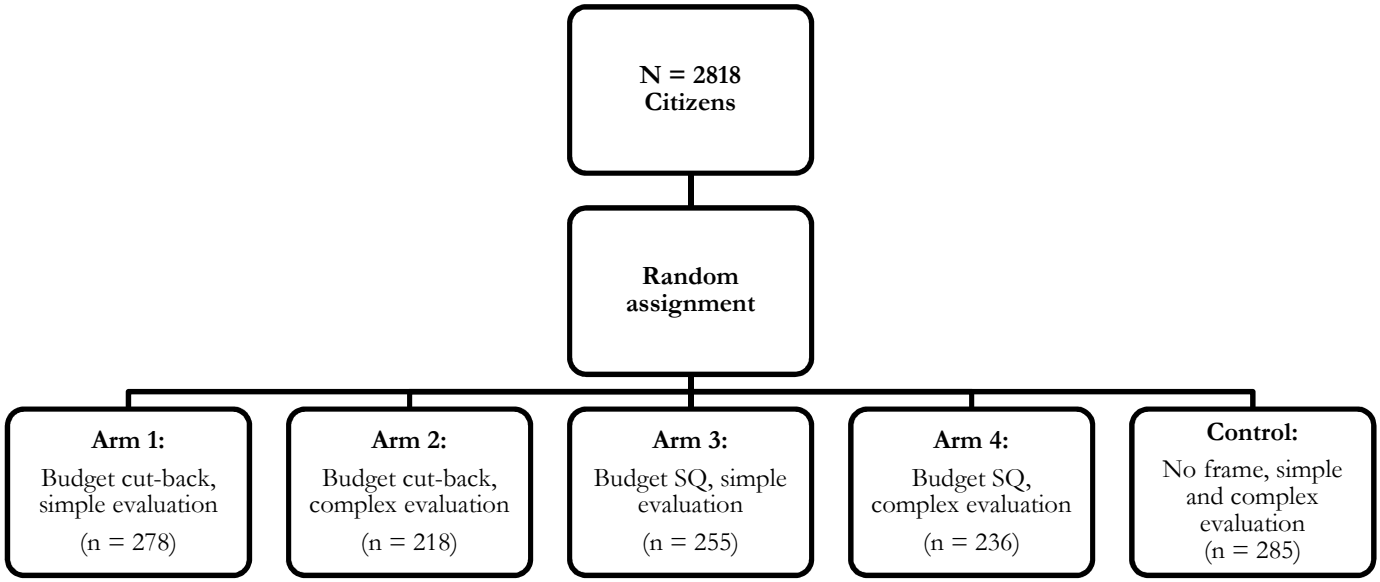
In Study 1, conducted in the fall of 2013, respondents were randomized into four different experimental conditions and a control group. The conditions were constructed from two different types of treatments. The first type was a priming of the respondents via two different text paragraphs from the budget of 2013 from the City of Aarhus, the City in which all of the respondents' children were attending public schools. Both of the short paragraphs were therefore real statements from the City's politicians. The paragraph in the first treatment emphasized that the City still was struggling with the effects of the economic crisis and therefore had to cut back on expenses. The paragraph from the second treatment emphasized that although the city as a whole was supposed to cut back on expenses, the public schools were to be exempted from this requirement.

The other treatment type varied between two different sets of questions that respondents were asked to answer in the survey. In the first set the respondents were asked to evaluate the reading materials at their child's school through three different questions. The first question was about their expectations toward the reading materials supplied at their child's school, the next about their perception of the quality of the reading materials, and the last question was about their satisfaction with the reading materials. These questions were constructed to evaluate a part of the service that is rather concrete and therefore should be fairly simple for the parents to assess.

The second set of questions were structured the same way as the first but asked the parents to evaluate the pedagogic efforts of the school in connection with teaching the children to read instead of the specific reading materials. These questions were constructed to evaluate a part of the service that is more abstract and therefore would be harder for the parents to assess.

The overall design, N and the response rate in the first survey experiment is illustrated in figure 2. For full question and frame wording see appendix A.

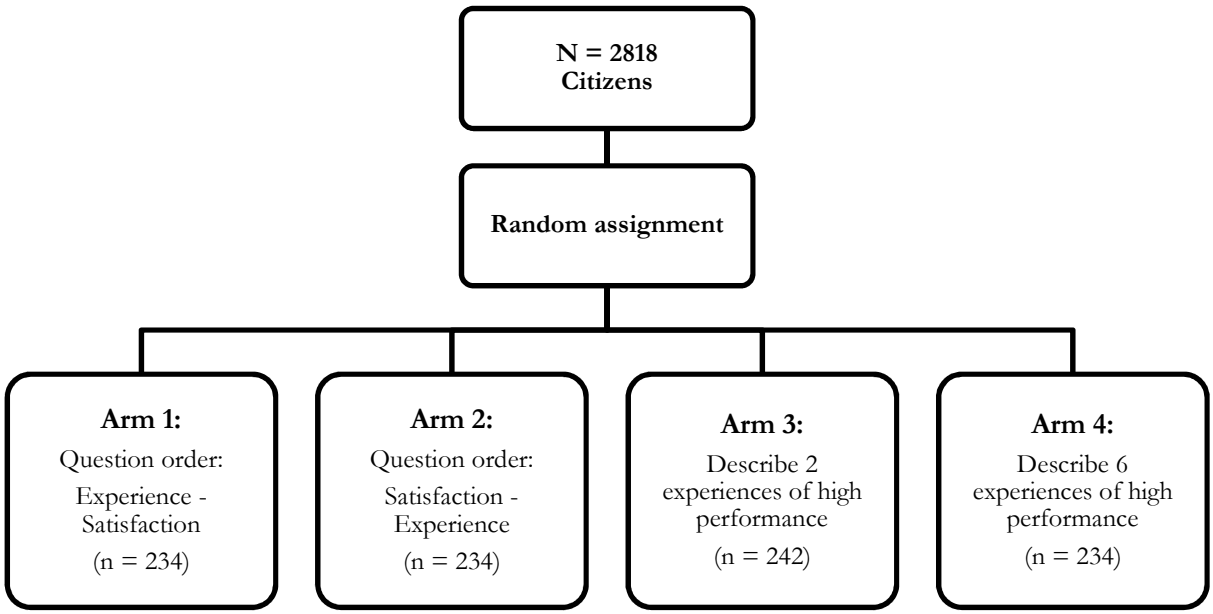
Figure 2: Design of Survey Experiment 1



Note: N denotes the number of citizens in the study population. n denotes the number of respondents who answered the survey

Study 2 was embedded in the survey sent out to the same parents in the spring of 2014. The treatments in this experiment were firstly the order of the questions about perceived performance and satisfaction. In both conditions the parents were (again) asked to evaluate the pedagogic efforts in connection with teaching the children to read. In the treatment condition, the question concerning the perceived performance of the pedagogic efforts was placed in front of the question about their satisfaction with these efforts. This was reversed in the control condition. Here, the question about the parents' satisfaction with the pedagogic efforts of the school in connection with the teaching of reading capabilities was placed first, and the question about the perceived performance followed directly after. Secondly, two further treatments asked the parents to think about 2 or 6 specific experiences in which they experienced high quality at their child's school and to shortly describe those experiences. These treatments, with the exception of the treatment with the satisfaction question first (arm 2), were designed to make the parents think of their experiences with the performance at their child's school before answering the satisfaction question. Figure 3 illustrates the design of survey experiment 2.

Figure 3: **Design of Survey Experiment 2**



Note: N denotes the number of citizens in the study population. n denotes the number of respondents who answered the survey

Data and Methodology

As mentioned, the two surveys were sent out to the parents of the children in 2nd and 3rd grade levels in the 29 schools participating in the study. We obtained the addresses, emails and basic information on the children and parents via central registers at The City of Aarhus. 2818 children were in the sample across the two surveys[3]. Survey 1 had a response rate of 45.1 pct., while Survey 2 had a response rate of 33.5 pct. meaning that 1272 and 944 of the childrens' parents answered the surveys, respectively[4].

The surveys were sent out via mail to one or both of the parents if there was a registered email address on either of them. If not, a paper version of the survey was sent out including information on how to fill out the survey on the internet. Two reminders were sent subsequently by email or direct

[3] Some children changed to schools not participating in the study, while others started at the participating schools during the study. The 2818 are those who were registered at the schools throughout the period.
[4] The response rate is calculated from the number of respondents who reached the end of the internet questionnaire or returned the questionnaire via direct mail. This does not necessarily mean that the respondents answered all questions.

mail. It was possible to answer the surveys in 5 different languages (Danish, English, Arabic, Somali and Turkish)[5] corresponding to the largest language groups in Aarhus. This approach attempted to reach as many parents in the target group as possible.

Two challenges emerged from the chosen sampling method: 1) Some children had answers from both of their parents and 2) some families with children in 2nd and 3rd grade have more than 1 child in this age cohort and would therefore receive more than one survey. Firstly, for the purpose of this study, mother's answer was chosen if both parents had answered the survey. Secondly, it was ensured that families with more children included in the study always had the same treatment if they had more than one survey. Moreover, the analysis uses clustered standard errors to account for the possibly high correlation between two or more[6] answers from the same family.

Apart from the surveys, register data from Statistics Denmark are utilized. These data are considered highly reliable and contain information on many background characteristics, e.g., income, number of children, unemployment, industry, full- or part time employment. The register data are connected to the survey data through the initial data received from The City of Aarhus. This also means that we have the background data on both the respondents who answered the survey and those who did not.

The measurement of satisfaction, expectation and perceived performance is three five-level likert items (see Appendix A). Since these are ranked, the normal procedure would be to use ordinal logit models when using a single item as the dependent variable. However, in our case the parallel regression assumption is broken and the ordinal logit models infeasible (Long and Freese 2014, pp.326–331)[7].

At the same time, the distribution of the items is highly left skewed (most respondents are either satisfied or very satisfied, have high expectations or evaluate the service as often performing at a

[5] The paper version sent out via direct mail was in Danish but included information in all 5 languages on how to find the survey on the internet.
[6] Of those who answered the surveys, only 1 family had 3 children in the first survey while 110 families had two children. 71 families had two children in the second survey, none had 3.
[7] Evaluated with the Brant test in Stata 13

high level). This is typical for the EDT items, where the extreme points on the scale are found to be of higher importance and constitute better performance indicators (Lyons *et al.* 1992; James 2009, p.112). As a consequence, we created dummies indicating "Satisfied" and "Very satisfied" for the satisfaction item and "Often" and "Very often" for the expectations and perceived performance items, and analyzed the experiments with logistic regression models.

IV. Results

The next two sections present the results from the two survey experiments. Firstly, it is considered whether budget frames can affect the citizens' evaluation of the performance of two different parts of a service even though it is an evaluation of personal experiences situated in the past, and therefore should not logically be changed by introducing information on the budget circumstances in a survey. The effect of the frames on expectation and satisfaction is also considered, since this would be in accordance with the EDT. Information on current and future budgets should be able to change the expectations of citizens about the service and therefore also – following the logic of EDT – the satisfaction.

Secondly, in Study 2 we investigate the possibility that the question of perceived performance itself can bias the answers to the satisfaction question, bringing the link between the perceived performance and satisfaction further into question. Moreover, it is considered if asking the citizens to shortly describe 2 or 6 experiences has a different effect on their satisfaction than the classic question of perceived performance. Asking the citizens to exhibit a little more effort in their evaluation could be a way of activating their system 2, which in turn could diminish the bias in their satisfaction evaluation.

But before turning to these results, it would be interesting to check if the classic test of the EDT works in our sample. If not, we would have a hard time claiming the generalizability of our results to

the classic tests of the theory, which typically find the posited relationship to be true (Van Ryzin 2004, 2006; James 2009; Morgeson 2012). The next section presents such a simple test of the EDT.

The Classic Test of the EDT

To get a good sense of the data, we start by investigating how the different variables correlate with each other. Table 1 andTable 2 display pearson's correlations between the three EDT items and the subtractive disconfirmation measure. The subtractive disconfirmation measure is a common way of constructing a measure of disconfirmation, where the expectation item is simply subtracted from the perceived performance measure creating a variable ranging from -4 to 4 indicating if the expectations exceed the perceived performance[8] (Van Ryzin 2006; James 2009).

Table 1 displays the correlations among the respondents who had the simple service question about the reading materials. Not surprisingly, the highest correlations are among the disconfirmation measure and the perceived performance and the expectation measures, since the disconfirmation measure is constructed from these variables. But the correlation between satisfaction and the perceived performance is also fairly high. Also, the expectations and the perceived performance variables have a medium correlation.

Table 1: Simple Question – Correlations Between Key EDT Variables

	Satisfaction	Perceived Performance	Expectations	Disconfirmation
Satisfaction	1			
Perceived Performance	0.457*	1		
Expectations	0.136*	0.389*	1	
Disconfirmation	0.277*	0.526*	-0.580*	1

Note: * $p < 0.01$

[8] Another common way is to directly ask the respondent about the confirmation or disconfirmation of expectations (the perceived disconfirmation, see Van Ryzin (2006))

The correlation between satisfaction and disconfirmation, on the other hand, does not seem that high. The same goes for the correlation between satisfaction and expectations, which is in accordance with previous research.

Table 2 lists the correlations from the same exercise on the more complex questions on the applied pedagogy in connection with the teaching of reading at the school. By and large, the picture is the same as with the simple questions. High correlations between the expectations, perceived performance and the disconfirmation variables, medium between satisfaction and perceived performance and a low correlation between the expectations and the satisfaction and the disconfirmation variables. If anything, the correlations seem slightly higher between the variables in the complex service evaluation case.

Table 2: Complex Question – Correlations Between Key EDT Variables

	Satisfaction	Perceived Performance	Expectations	Disconfirmation
Satisfaction	1			
Perceived Performance	0.506*	1		
Expectations	0.186*	0.350*	1	
Disconfirmation	0.349*	0.681*	-0.448*	1

Note: * $p < 0.01$

Table 3 contains logistic regressions with the three components of the EDT on the citizens' satisfaction (1 = "Satisfied" or "Very Satisfied", 0 otherwise). Although the usual test is done with structural equation models (e.g., Van Ryzin 2004, 2006; Van Ryzin *et al.* 2004; Morgeson 2012), we investigate the EDT only with the expectation and perceived performance or the disconfirmation constructs in the models. Prior research has often included both the two individual items (expectations and perceived performance) and the disconfirmation construct (e.g., Van Ryzin 2004, 2006) in the same model, but to avoid perfect multicollinearity this practice requires at least two different measures of one

of the two individual items, which we do not have[9]. One way to circumvent this problem is to only include one of the two individual items in the regression (e.g., James 2009), but this also has the downsides of "high" but "imperfect" multicollinearity (Gujarati 2009 Ch. 10).

Table 3: Classic Investigation of the EDT's Predictions (Pooled Data)

	Expectations and Perceived Performance		Disconfirmation	
	(1)	**(2)**	**(3)**	**(4)**
Expectations	0.878	0.852		
	(0.109)	(0.110)		
Perceived Performance	5.241***	5.535***		
	(0.849)	(0.931)		
Disconfirmation			2.409***	2.495***
			(0.278)	(0.297)
Constant	0.021***	0.017**	9.828***	10.112
	(0.0129)	(0.0280)	(1.384)	(15.34)
Controls	No	Yes	No	Yes
N	1171	1171	1171	1171

Note: * p < 0.10, ** p < 0.05, *** p < 0.01. Two-tailed tests. Logit models with standard errors clustered at the family level. Exponentiated coefficients (odds ratios); Standard errors in parentheses.

Therefore, model 1 and 2 in Table 3 only includes the expectation and the perceived performance items (with and without controls). We have pooled the two questions (simple and complex evaluations) and treatment and control groups here, but the results can be reproduced in each of the groups individually with basically the same results. As also indicated by the simple correlations in Table 1 and Table 2, the expectations item has no direct effect on satisfaction when we control for perceived performance. On the contrary, perceived performance has a statistically significant, and large, effect on satisfaction when holding expectations constant. With odds ratios of above 5 this effect means that a

[9] One example is to have a single item measure of the experienced performance in the survey, which can be used to create the subtractive disconfirmation measure, and then use multiple items from different service areas to create a single latent measure of performance to be used in the structural equation model. This way the latent measure in the model will not perfectly predict the subtractive disconfirmation measure (Van Ryzin 2004, 2006).

citizen has above 5 times as high a probability of being "Satisfied" or "Very Satisfied" if she moves 1 scale point on the perceived performance item[10].

The effect of the subtractive disconfirmation construct on satisfaction is illustrated in model 3 and 4 in Table 3. Here we also see a clear statistically significant effect on satisfaction indicating that the correlation expected by EDT is found in our sample. A one point increase on the subtractive disconfirmation scale (scaled from -4 to 4) results in a 2.41 and 2.50 times as high a probability for being "Satisfied" or "Very Satisfied"[11].

Study 1: Can Frames Affect the Perceived Performance and Expectations?

In the first experiment we test how two different frames about the budget situation in the City of Aarhus will affect the EDT's basic assumptions. Table 4 lists the results on the logistic regressions of being "Satisfied" or "Very Satisfied" (the satisfaction item) and "Often" or "Always" (the perceived performance and the expectations items). The two budget frames are both word-by-word descriptions taken from the actual budget of The City of Aarhus[12]. We first look at the treatment telling citizens that budget constraints will results in cut backs. As can be seen from Model 1 this treatment negatively affects the perception of the experiences with the more abstract performance (i.e. the pedagogy used in their kids' school) – even though the budget situation should not affect the experience of past performance. Both frames have a somewhat equal negative effect on the perceived performance. Receiving the budget cut-back frame means that the probability of evaluate the experience as high performance "Often" or "Always" is halved as compared with the control group. This seems to suggest that this priming is not used as content information, but makes parents associate budget constraints with performance and therefore respond more negatively to the performance experience question.

[10] Speaking in terms of a one standard deviation increase on the experienced performance scale, the odds of being satisfied or very satisfied are 3.63 and 3.78 times greater in model 1 and 2 respectively.
[11] In standard deviations: 2.08 and 2.14 times greater odds of being satisfied or very dissatisfied
[12] The City of Aarhus had to generally cut back on expenses in 2013, but would not do so on schools, where The City was able to uphold a status quo. For exact wording see Appendix A

To further demonstrate that the content information is not used, we look at the other priming of the citizens, which stated that despite the budget constraints there would be no cut backs on the school area (status quo). The result shows (Model 1, second row) that this has just as negative effect on parents' response about their perceived performance as the cut-back priming.[13] This supports the interpretation that citizens' respond in a way that use an associative mode of thinking (System 1) rather than a more conscious mode (System 2) that should disregard information about future situations that are irrelevant to past performance.

Next, it could be assumed that citizens' responses are more easily influenced at more abstract performance areas, which they may not have much of an idea about (as in the general performance of a local authority in James' (2011) experiment). However, we find the same effect when we ask about the experience with a more concrete service area (reading materials such as books) (Model 4). Again this supports the notion that an associative mode is driving these effects rather than more reflective ways of evaluating past performance.

Table 4: Can Frames Affect the Perceived Performance and Expectations?

	Abstract Judgement			Concrete Judgement		
	Perceived Performance	**Expectations**	**Satisfaction**	**Perceived Performance**	**Expectations**	**Satisfaction**
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Cutback	0.406***	0.570	0.483***	0.501***	0.862	0.400***
	(0.107)	(0.217)	(0.121)	(0.0989)	(0.224)	(0.104)
Status Quo	0.532**	1.388	0.587**	0.467***	0.689	0.349***
	(0.144)	(0.681)	(0.150)	(0.0938)	(0.175)	(0.0911)
Constant	7.769***	20.54***	7.437***	4.036***	7.324***	10.36***
	(1.625)	(5.845)	(1.400)	(0.604)	(1.378)	(2.176)
N	680	732	722	814	815	816

Note: * p < 0.10, ** p < 0.05, *** p < 0.01. Two-tailed tests. Logit models with standard errors clustered at the family level. Exponentiated coefficients (odds ratios); Standard errors in parentheses.

[13] There is no statistically significant difference between the two point estimates, which goes for all the significant point estimates in models 1-6 in table 4

The frames do neither affect the expectations in the abstract judgment case nor in the concrete judgment case (model 2 and 5). This is surprising, because according to the EDT, expectations, if anything, should be amendable to new information about the state of the world (James 2011b). Satisfaction is, on the other hand, affected negatively by both frames, which is in accordance with the drop in perceived performance (models 3 and 6).

Based on these findings we conducted a second study to further explore whether the relationship between perceived performance and satisfaction – a core relationship in the EDT – is also subject to an associative mode of thinking.

This pattern repeats itself in models 4-6 where citizens are asked to evaluate a more complex service. There are negative effects from both frames on experienced performance and satisfaction, but not on expectations. Again, there are no significant differences between the effects of the frames. Neither are there significant different effects between the two evaluation types in general. It seems that the citizens are affected by the frames in the same way across these questions.

Study 2: Perceived Performance as Primer?

In survey experiment 2 we look further into the relationship between the perceived performance and satisfaction constructs of the EDT. Firstly, we look at the possible priming effects of the typical perceived performance item on the satisfaction item by changing the order of the two items experimentally. This is done to test if the citizens substitute the question about satisfaction with an easier question about the primed information (System 1), that is, information from the perceived performance question. (If so, this would be information that is susceptible for malleability by irrelevant information as we have seen in Study 1).

Secondly, we test an alternative way of having the citizens think about their experiences with the service – a more specific and effortful way that might activate their system 2 way of thinking instead of the more effortless and associative System 1 way of thinking – and see if this has an effect on their

satisfaction. Specifically, we ask the citizens to describe 2 or 6 instances where they experience a high performance at their child's school.

This is done to test if the citizens rely on the ease of retrieval rather than the content information, which would manifest itself in a difference in satisfaction between those who were asked to describe 2 and those who were asked to describe 6 instances during the last week. Furthermore, it enables us to test if citizens react differently when primed with another kind of perceived performance question that encourages them to activate system 2.

In both treatment types (priming with the perceived performance question and asking for 2 or 6 descriptions), we subsequently asked citizens to evaluate the applied pedagogy in connection with the teaching of reading at their childrens' schools (the abstract judgement, see Appendix A).

Figure 4 shows the distribution of the 4 different treatments in terms of the citizens' satisfaction. All four distributions are left skewed as is often the case with satisfaction questions. There are, however, differences between the distributions. The difference between the two first treatments primarily occurs between the "Neither/Nor" and the "Satisfied" answers, when comparing the results from having the satisfaction question first (Satisfaction-Experience) to the case where citizens had the experience question first (Experience-Satsifaction). Here, citizens seem somewhat more satisfied when first asked about their perceived performance evaluation than if not. This points to a possible priming effect of the experience question.

On the other hand, there is no apparent difference between the two different description treatments. It appears to make no difference to the satisfaction if the citizen is asked to describe either 2 or 6 instances in which they experienced high performance. However, in comparison with the other perceived performance priming, there is a difference in how the citizens react in terms of satisfaction. They are more satisfied in the Experience-Satisfaction condition than in the other conditions.

Figure 4: Distribution of Satisfaction across Treatment and Control Groups

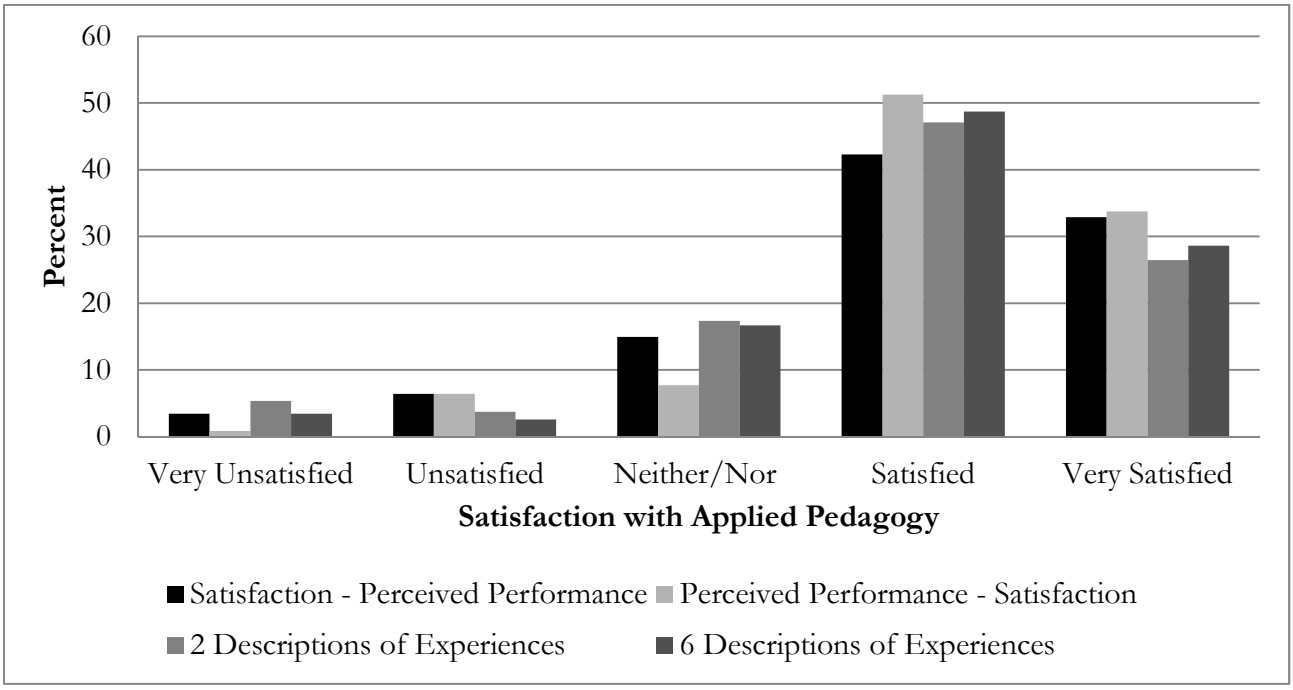


Table 5 tests the difference between the first two treatments in Study 2: Priming of satisfaction with the perceived performance question and no priming, with and without controls. The dependent variable is the satisfaction dummy with "Satisfied" and "Very Satisfied" answers as 1 and the rest as 0. It turns out that the difference between having the standard perceived performance question before answering the satisfaction question is significant. The citizens in the group which had the experience primer first are 77 pct. more likely to be satisfied than the group not having the experience question as primer. Introducing the controls pushes the likelihood up to 88 pct.

This means that the standard question of perceived performance indeed has an effect on the citizens' satisfaction. Possibly, they rely on the easy retrieval of the information that the perceived performance question create and use this when they evaluate the satisfaction. However, this turns out on average to be a more positive evaluation than if the perceived performance question is not asked first.

Table 5: Does the Perceived Performance Question Work as a Primer on Satisfaction?

	(1)	(2)
Priming	1.769**	1.878**
	(0.496)	(0.579)
No Priming	Ref.	Ref.
	-	-
Constant	3.605***	121.8***
	(0.625)	(218.1)
Controls	No	Yes
N	375	375

Note: * p < 0.10, ** p < 0.05, *** p < 0.01. Two-tailed tests. Logit models with standard errors clustered at the family level. Exponentiated coefficients (odds ratios); Standard errors in parentheses.

Table 6 shows two logit models testing the difference between the two "description" treatments in terms of satisfaction. As is evident, there is no significant difference between being asked to describe either two or six instances during the last week where the citizens experienced a high service quality in terms of satisfaction. This can also be seen from Figure 4 in which there are only very small differences between these groups.

Table 6: Difference between the Two Description Treatments

	(1)	(2)
2 Descriptions	0.951	1.044
	(0.244)	(0.299)
6 Descriptions	Ref.	Ref.
	-	-
Constant	3.381***	0.673
	(0.606)	(1.453)
Controls	No	Yes
N	361	361

Note: * p < 0.10, ** p < 0.05, *** p < 0.01. Two-tailed tests. Logit models with standard errors clustered at the family level. Exponentiated coefficients (odds ratios); Standard errors in parentheses.

Apparently, there is no "ease of retrieval" effect here (Schwarz *et al.* 1991). However, it is also interesting how the different ways of asking the citizens about their perceived performance compare. Table 7 compares the description treatments to the Satisfaction-Experiences and Experiences-Satisfaction treatments respectively.

Table 7: Comparison of All Four Treatments

	(1)	**(2)**
Experiences - Satisfaction	1.864***	Ref.
	(0.451)	-
Satisfaction - Experiences	Ref.	0.536***
	-	(0.130)
2 Specific Experiences	0.917	0.492***
	(0.195)	(0.118)
6 Specific Experiences	1.125	0.604**
	(0.247)	(0.149)
Constant	3.034***	5.657***
	(0.460)	(1.065)
N	943	943

Note: * p < 0.10, ** p < 0.05, *** p < 0.01. Two-tailed tests. Logit models with standard errors clustered at the family level. Exponentiated coefficients (odds ratios); Standard errors in parentheses.

When compared to the case where the citizens first had the satisfaction question (without priming) in model 1, the description treatments do not have a significant impact on satisfaction. Therefore, asking citizens to think about specific instances and describing them before answering the satisfaction question does not change their satisfaction as compared with those, who were not asked to describe their experiences.

On the other hand, there are significant effects from both description treatments in model 2 where they are compared to the experimental group, which had the classic experiences question before answering the satisfaction question. Being asked to describe 2 instances make the odds of being satisfied decrease by a factor of 0.49, meaning that citizens in this group were half as likely to be "Satisfied" or "Very Satisfied" as the citizens in the Experiences-Satisfaction treatment. Likewise, being asked to

describe 6 instances of high quality decreases the odds of being satisfied by a factor of 0.60. The difference between the two description treatments' effects is not significant.

This result indicates that other ways of asking about citizens about their perceived performance might not bias their answers like the standard question does.

V. Conclusion

[Yet to be completed]

Citizen satisfaction surveys are used for both research and management either as a measure of performance or to monitor if performance changes generate an increase in citizens' satisfaction with the services. The dominating theory in this field, the Expectations-Disconfirmation Theory (EDT) suggest that there should be a computationally rational releationship between performance, perceived performance and satisfaction (for any given level of expectations). However, what we show in the studies presented here, is that this relationship is dominated by a much less rational relationship – a relationship that can be interpreted as a result of a more associative mode of thinking.

We do not claim that citizens' satisfaction is completely unrelated to performance and expetations. However, evidence for EDT is not so strong given that only few studies test the endogeneous model experimentally, and one of those that does, do not find the hypothesized relationship between expectations and satisfaction (Van Ryzin 2013). Further, the research we have presented here shows that the relationship is clearly not as simple as suggested by EDT.

These findings lay the stepping stones for a more comprehensive theory of citizen satisfaction. But they also indicate ways of measuring satisfaction that are less prone to irrelevant associations. One way could be to use separate data sources for expectations, performance and satisfaction as for instance panel studies. Another way could build on the finding that satisfaction remains unchanged when citizens were asked to remember a number of specific experiences of high quality. It could seem as if

this exercise makes citizens activate their more reflective thinking (System 2) and therefore reduces the influence of the performance question on the satisfaction question.

VI. Literature

Bouckaert, G., Walle, S.V. de, Kampen, J.K. (2005) 'Potential for comparative public opinion research in public administration', *International Review of Administrative Sciences*, 71(2), 229–240.

Cardozo, R.N. (1965) 'An experimental study of customer effort, expectation, and satisfaction', *Journal of Marketing Research*, 244–249.

Charbonneau, É., Van Ryzin, G.G. (2012) 'Performance Measures and Parental Satisfaction With New York City Schools', *The American Review of Public Administration*, 42(1), 54–65.

Greifeneder, R., Bless, H., Pham, M.T. (2011) 'When Do People Rely on Affective and Cognitive Feelings in Judgment? A Review', *Personality and Social Psychology Review*, 15(2), 107–141.

Gujarati, D.N. (2009) *Basic Econometrics*, 5 edition. ed, McGraw-Hill Higher Education: Boston.

Helson, H. (1948) 'Adaptation-level as a basis for a quantitative theory of frames of reference', *Psychological Review*, 55(6), 297–313.

Imai, K., Keele, L., Tingley, D., Yamamoto, T. (2011) 'Unpacking the Black Box of Causality: Learning about Causal Mechanisms from Experimental and Observational Studies', *American Political Science Review*, 105(04), 765–789.

James, O. (2009) 'Evaluating the Expectations Disconfirmation and Expectations Anchoring Approaches to Citizen Satisfaction with Local Public Services', *Journal of Public Administration Research and Theory*, 19(1), 107.

James, O. (2011a) 'Performance Measures and Democracy: Information Effects on Citizens in Field and Laboratory Experiments', *Journal of Public Administration Research and Theory*, 21(3), 399–418.

James, O. (2011b) 'Managing Citizens' Expectations of Public Service Performance: Evidence from Observation and Experimentation in Local Government', *Public Administration*, 89(4), 1419–1435.

Kahneman, D. (2011) *Thinking, Fast and Slow*, 1 edition. ed, Farrar, Straus and Giroux: New York.

Kahneman, D., Frederick, S. (2002) 'Representativeness revisited: Attribute substitution in intuitive judgment', *Heuristics and biases: The psychology of intuitive judgment*, 49–81.

Kelly, J.M., Swindell, D. (2003) 'The Case For The Inexperienced User: Rethinking Filter Questions in Citizen Satisfaction Surveys', *The American Review of Public Administration*, 33(1), 91–108.

Long, J.S., Freese, J. (2014) *Regression Models for Categorical Dependent Variables Using Stata, Third Edition*, 3 edition. ed, Stata Press: College Station, TX.

Lyons, W.E., Lowery, D., DeHoog, R.H. (1992) *The Politics of Dissatisfaction: Citizens, Services, and Urban Institutions*, ME Sharpe.

Miller, T.I., Kobayashi, M.M., Hayden, S.E. (2009) *Citizen Surveys for Local Government*, 3rd ed, International City Management Association.

Morgeson, F.V. (2012) 'Expectations, Disconfirmation, and Citizen Satisfaction with the US Federal Government: Testing and Expanding the Model', *Journal of Public Administration Research and Theory*, Online Publication April 19th 2012.

Mutz, D.C. (2011) *Population-Based Survey Experiments*, Princeton University Press: Princeton.

Oliver, R.L. (1980) 'A cognitive model of the antecedents and consequences of satisfaction decisions', *Journal of marketing research*, 17(4), 460–469.

Oliver, R.L. (1997) *Satisfaction: A Behavioral Perspective on the Customer*, McGraw-Hill: Boston.

Oliver, R.L., DeSarbo, W.S. (1988) 'Response determinants in satisfaction judgments', *Journal of consumer research*, 495–507.

Olshavsky, R.W., Miller, J.A. (1972) 'Consumer expectations, product performance, and perceived product quality', *Journal of Marketing Research*, 19–21.

Olson, J.C., Dover, P. (1976) 'Effects of expectation creation and disconfirmation on belief elements of cognitive structure', *Advances in consumer research*, 3, 168–175.

Parks, R.B. (1984) 'Linking objective and subjective measures of performance', *Public Administration Review*, 44(2), 118–127.
Poister, T.H., Thomas, J.C. (2011) 'The Effect of Expectations and Expectancy Confirmation/Disconfirmation on Motorists' Satisfaction with State Highways', *Journal of Public Administration Research and Theory*, 21(4), 601–617.
Van Raaij, W.F. (1989) 'Economic news, expectations and macro-economic behaviour', *Journal of Economic Psychology*, 10(4), 473–493.
Roch, C.H., Poister, T.H. (2006) 'Citizens, Accountability, and Service Satisfaction: The Influence of Expectations', *Urban Affairs Review*, 41(3), 292–308.
Van Ryzin, G.G. (2004) 'Expectations, performance, and citizen satisfaction with urban services', *Journal of Policy Analysis and Management*, 23(3), 433–448.
Van Ryzin, G.G. (2006) 'Testing the expectancy disconfirmation model of citizen satisfaction with local government', *Journal of Public Administration Research and Theory*, 16(4), 599.
Van Ryzin, G.G. (2013) 'Experimental Test of the Expectancy-Disconfirmation Theory of Citizen Satisfaction', *Journal of Policy Analysis and Management*, 32, 597–614.
Van Ryzin, G.G., Muzzio, D., Immerwahr, S., Gulick, L., Martinez, E. (2004) 'Drivers and consequences of citizen satisfaction: An application of the American Customer Satisfaction Index model to New York City', *Public Administration Review*, 64(3), 331–341.
Schwarz, N., Bless, H., Strack, F., Klumpp, G., Rittenauer-Schatka, H., Simons, A. (1991) 'Ease of retrieval as information: Another look at the availability heuristic.', *Journal of Personality and Social psychology*, 61(2), 195.
Schwarz, N., Clore, G.L. (2007) 'Feelings and Phenomenal Experiences', in Higgins, E.E. and Kruglanski, A., eds., *Social Psychology: Handbook of Basic Principles*, Guilford Press, 385–407.
Stanovich, K.E., West, R.F. (2000) 'Individual differences in reasoning: implications for the rationality debate?', *The Behavioral and Brain Sciences*, 23(5), 645–665; discussion 665–726.
Stipak, B. (1979) 'Citizen satisfaction with urban services: Potential misuse as a performance indicator', *Public Administration Review*, 39(1), 46–52.
Strack, F., Martin, L., Schwarz, N. (1988) 'Priming and Communication - Social Determinants of Information Use in Judgments of Life Satisfaction', *European Journal of Social Psychology*, 18(5), 429–442.
Tse, D.K., Wilton, P.C. (1988) 'Models of consumer satisfaction formation: an extension', *Journal of marketing research*, 25(2), 204–212.
Tversky, A., Kahneman, D. (1974) 'Judgment under uncertainty: Heuristics and biases', *science*, 185(4157), 1124–1131.
Van de Walle, S., Van Ryzin, G.G. (2011) 'The Order of Questions in a Survey on Citizen Satisfaction with Public Services: Lessons from a Split-ballot Experiment', *Public Administration*, available: http://onlinelibrary.wiley.com/doi/10.1111/j.1467-9299.2011.01922.x/abstract [accessed 15 Apr 2011].
Yi, Y. (1990) 'A Critical Review of Consumer Satisfaction', in *Review of Marketing 1990*, Marketing Classics Press: University of Pittsburgh, 68–123, available: http://www.google.dk/books?hl=da&lr=&id=4TUXGG0Cf5cC&oi=fnd&pg=PA68&ots=VnkkeO_JLs&sig=3cvGoR7uGt6q6PQjK8zPmN2vQM0&redir_esc=y#v=onepage&q&f=false.
Zeithaml, V.A., Berry, L.L., Parasuraman, A. (1993) 'The nature and determinants of customer expectations of service', *Journal of the academy of Marketing Science*, 21(1), 1–12.

VII. Appendix A – Frames and Questions from the Surveys

Study 1

Experimental condition	Frame	Questions		
		1	2	3
Arm 1	Budget cut-back	How often do you expect the reading materials (fx. books) at your child's school to be of high quality?	How often do you experience the reading materials (fx. books) at your child's school to be of high quality?	How satisfied are you with the reading materials at your child's school?
Arm 2	Budget cut-back	How often do you expect the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?	How often do you experience the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?	How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?
Arm 3	Budget status quo	How often do you expect the reading materials (fx. books) at your child's school to be of high quality?	How often do you experience the reading materials (fx. books) at your child's school to be of high quality?	How satisfied are you with the reading materials at your child's school?
Arm 4	Budget status quo	How often do you expect the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?	How often do you experience the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?	How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?
Control	None	How often do you expect the reading materials (fx. books) at your child's school to be of high quality? How often do you experience the reading materials (fx. books) at your child's school to be of high quality?	How often do you expect the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality? How often do you expect the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?	How satisfied are you with the reading materials at your child's school? How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?

Note: The questions in the control condition were asked in this order: Expect/experience reading materials, expect/experience pedagogy, satisfaction with reading materials, satisfaction with pedagogy

Budget cutback frame

Schools have to work under a set of different economic constraints. The politicians in the City of Aarhus have decided the following about the economy in 2013:

"The city's economy is, as the rest of the country, affected by the economic crisis. The reforms of the employment subsidy and the equalization systems in the spring – together with the agreement between KL and the government on local finances in 2013 - resulted in significantly less resources than expected. The external changes have meant that it is necessary to take measures to sustain the economic development and the settlement of debt, which was agreed last year.
The conciliation parties emphasize the continued control of the municipality's financial situation. As part of this, it is necessary to reduce the expenditures of the City."

Budget status quo frame

Schools have to work under a set of different economic constraints. The politicians in the City of Aarhus have decided the following about the economy in 2013:

"The conciliation parties place great emphasis on creating stability in the large service sectors of the Children and Youth Department in respect for the children, parents and employees. This applies for daycare institutions, schools, after school arrangements and youth clubs where the conciliation parties wishes to uphold the current quality.
The conciliation parties at the same time agree that the general savings on 1,4% will be found without afflicting the existing service quality including the planned number of teaching hours."

Study 2

Experimental condition	Questions	
Arm 1	How often do you experience the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?	How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?
Arm 2	How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?	How often do you experience the applied pedagogy in connection with the teaching of reading at your child's school to be of high quality?
Arm 3	In the next question, we will ask you to think about 2 situations during the last week where you have experienced a high quality in your child's school. Shortly describe the 2 situations:	How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?
Arm 4	In the next question, we will ask you to think about 6 situations during the last week where you have experienced a high quality in your child's school. Shortly describe the 6 situations:	How satisfied are you with the applied pedagogy in connection with the teaching of reading at your child's school?

VIII. Appendix B – Balance

Study 1

Table 1: Means

Variable	Arm 1	Arm 2	Arm 3	Arm 4	Control
Child's age	4.55	4.51	4.52	4.51	4.54
Mother's age	36.46	36.66	36.26	36.49	36.15
Child's gender	0.53	0.50	0.49	0.50	0.49
Mother is employed	0.86	0.87	0.84	0.86	0.86
Married	0.76	0.73	0.73	0.80	0.73
Disposable income	261,518.50	258,759.00	257,639.50	281,346.90	263,665.20
Number of children	2.22	2.20	2.21	2.29	2.29
Employment hours					
Not stated	0.21	0.19	0.21	0.21	0.22
Less than 32 hours	0.24	0.16	0.21	0.17	0.19
More than 32 hours	0.55	0.65	0.59	0.62	0.59
Mother's education					
None stated	0.03	0.03	0.04	0.04	0.04
Low	0.09	0.08	0.09	0.06	0.07
Middle	0.31	0.24	0.26	0.19	0.23
High	0.57	0.65	0.62	0.71	0.66

Study 2

Table 1: Means

Variable	Arm 1	Arm 2	Arm 3	Arm 4
Child's age	4.46	4.51	4.60	4.50
Mother's age	36.21	35.95	35.63	36.44
Child's gender	0.44	0.57	0.49	0.50
Mother is employed	0.83	0.84	0.87	0.87
Married	0.72	0.74	0.74	0.77
Disposable income	261,871.80	258,616.50	258,616.50	265,265.80
Number of children	2.24	2.25	2.18	2.25
Employment hours				
Not stated	0.25	0.25	0.20	0.19
Less than 32 hours	0.16	0.20	0.18	0.19
More than 32 hours	0.59	0.54	0.62	0.62
Mother's education				
None stated	0.05	0.01	0.03	0.04
Low	0.12	0.11	0.08	0.07
Middle	0.27	0.26	0.26	0.30
High	0.56	0.61	0.63	0.60